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Chapter 7

Advanced Router Topics

Chapter 8 presents questions on some fun topics for the Cisco ICND1 (CCENT) certification exam. This chapter introduces questions related to configuring DNS name resolution, DHCP services, and network address translation. It is important to note that the new version of the ICND1 exam includes objectives on password recovery procedures and access lists.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Resolving hostnames
- Configuring DHCP service
- Implementing NAT
- Recovering passwords
- Configuring access lists (ACLs)

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- You can resolve names you use on the router either by viewing the hostname table on the router or by querying DNS. Use the `IP Host` command to add a hostname to the hostname table on the router. To view the hostname table, use the `show hosts` command.
- To configure DHCP on your router, you need to use the `IP dhcp pool` command to create an address pool. After you create the address pool, you specify the network range of addresses to give out with the `network` command.
- There are two types of NAT – static NAT and overloading. Static NAT associates a single public address to a single private address, while overloading is the concept that all private addresses translate to the one public address.
- Access lists are used to control the traffic entering or leaving the network. To filter by the source IP address of a packet, you should configure standard access lists; to filter by the source and destination IP address, source and destination port, and protocol, on the other hand, use extended access lists.

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Hostname Resolution

238–245 Choose the best answer(s).

238. You try to ping the Boston router and get the error shown in the figure below. What should you do to allow the ping to work?

```
NY-R1>ping BOS-R1
Translating "BOS-R1"...domain server (255.255.255.255)
% Unrecognized host or address or protocol not running.
```

(A) Add the entry to the NAT translation table

(B) Add the entry to the hostname table

(C) Add the entry to the MAC address table

(D) Change the encapsulation protocol

239. What command displays the hostname table on a router?

(A) `show ip hosts`

(B) `show ip names`

(C) `show names`

(D) `show hosts`

240. To configure your router to query a DNS server, what command do you use?

(A) `ip name-server 23.0.0.10`

(B) `name-server 23.0.0.10`

(C) `ip dns 23.0.0.10`

(D) `ip domain 23.0.0.10`

241. You have decided to remove the entry for the Boston router from your hostname table. What command would you use?

(A) NY-R1(config)#`no ip host BOS-R1`

(B) NY-R1#`no ip host BOS-R1`

(C) NY-R1#`no host BOS-R1`

(D) NY-R1(config)#`no host BOS-R1`

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242. What command would you use to resolve the BOS-R1 router to the IP address of 15.10.0.5?

(A) `host BOS-R1 15.10.0.5`

(B) `ip BOS-R1 15.10.0.5`

(C) `ip name BOS-R1 15.10.0.5`

(D) `ip host BOS-R1 15.10.0.5`

243. Which command has created the output shown in the figure below?

```
Default Domain is not set
Name/address lookup uses domain service
Name servers are 255.255.255.255

Codes: UN - unknown, EX - expired, OK - OK, ?? - revalidate
       temp - temporary, perm - permanent
       NA - Not Applicable None - Not defined

Host      Port  Flags      Age Type   Address(es)
BOS-R1    None  (perm, OK)  0   IP    24.0.0.2
NY-R1#
```

(A) `show systems`

(B) `show hosts`

(C) `show ip dhcp`

(D) `show ip arp`

244. When administering your Cisco router, it tries to do a DNS query every time you have a typo in a command. How can you disable this?

(A) `ip domain lookup`

(B) `no ip domain lookup`

(C) `no hosts lookup`

(D) `hosts lookup disabled`

245. You have configured your router for a name server, but you still find that names are not being resolved. What command would you execute?

(A) `ip domain-lookup`

(B) `no ip domain-lookup`

(C) `ip routing`

(D) `ip name server-lookup`

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Configuration of DHCP Service

246–252 Choose the best answer(s).

246. Which of the following DHCP commands would you use to specify that the address pool is to configure the default gateway on client computers?

(A) NY-R1(dhcp-config)#`default-gateway 23.0.0.1`

(B) NY-R1(dhcp-config)#`default-router 23.0.0.1`

(C) NY-R1#`default-router 23.0.0.1`

(D) NY-R1#`default-gateway 23.0.0.1`

247. Clients are no longer receiving IP addresses from the DHCP server running on your Cisco router. You suspect the DHCP service may have been shut down. How can you start it?

(A) `start dhcp`

(B) `net start dhcp`

(C) `ip start dhcp`

(D) `service dhcp`

248. You have configured the DHCP service on your router as a backup DHCP service. You wish to stop the DHCP service so that addresses are not being handed out till you need the service. What command would you use to stop the service?

(A) `no dhcp`

(B) `no service dhcp`

(C) `no dhcp service`

(D) `no services-dhcp`

249. You want to create configure DHCP on your router to give out IP addresses for the second subnet of 192.168.3.0/27. The addresses should be leased to clients for 7 days. What commands would you use?

(A) Use the following:

NY-R1(config)#`ip dhcp pool NY_Network`

NY-R1(dhcp-config)#`network 192.168.3.64 255.255.255.224`

NY-R1(dhcp-config)#`lease 7 0 0`

(B) Use the following:

NY-R1(config)#`dhcp pool`

NY-R1(dhcp-config)#`network 192.168.3.32 255.255.255.224`

NY-R1(dhcp-config)#`lease 7 0 0`

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(C) Use the following:

NY-R1(config)#`ip dhcp pool NY_Network`

NY-R1(dhcp-config)#`network 192.168.3.32 255.255.255.224`

NY-R1(dhcp-config)#`lease 7 0 0`

(D) Use the following:

NY-R1(config)#`ip dhcp pool NY_Network`

NY-R1(dhcp-config)#`lease 0 7 0`

250. You are monitoring DHCP usage on your router. What command would you use to see how many DHCP related messages the router has received?

(A) `show dhcp`

(B) `show ip dhcp server statistics`

(C) `show dhcp statistics`

(D) `show ip dhcp`

251. You are administering the DHCP service on your Cisco router and would like to look at the DHCP leases, What command would you use?

(A) `show ip dhcp leases`

(B) `show dhcp`

(C) `show ip dhcp binding`

(D) `show dhcp leases`

252. To view a list of IP addresses given to clients on the network by your Cisco router DHCP service, what command do you use?

(A) `show binding`

(B) `show ip dhcp binding`

(C) `show ip translations`

(D) `show ip dhcp translations`

Implementing NAT

253–262 Choose the best answer(s).

253. Which of the following implements NAT overload?

(A) FAT

(B) SNAT

(C) PAT

(D) JAT

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254. Your manager has asked what would be a good use of static NAT.

(A) Share a single public IP address to all internal systems

(B) For internal clients to surf the Internet

(C) To allow DHCP to assign the address

(D) For publishing an internal system to the Internet

255. You wish to create an access list to be used for NAT as the inside addresses that will be permitted to use NAT. What command would you use to create the access list?

(A) NY-R1(config)#`Access-list 16 permit 10.0.0.0 0.255.255.255`

(B) NY-R1(config)#`Access-list 16 permit 10.0.0.0 255.0.0.0`

(C) NY-R1`>Access-list 16 permit 10.0.0.0 255.0.0.0`

(D) NY-R1`>Access-list 16 permit 10.0.0.0 0.255.255.255`

256. Your serial port is connected to the WAN environment, with your Fast Ethernet port connected to the LAN. When configuring NAT, how would you configure the serial port?

(A) NY-R1(config-if)#`ip nat inside`

(B) NY-R1(config)#`ip nat inside`

(C) NY-R1(config)#`ip nat outside`

(D) NY-R1(config-`if)#ip nat outside`

257. What mechanism does NAT overloading use to allow one public IP address to be mapped to multiple systems?

(A) NAP

(B) TAM

(C) NAC

(D) PAT

258. You have created access list 1 that lists the IP addresses for NAT usage. What command would you use to configure the router so that those addresses can use NAT?

(A) NY-R1(config)#`nat access list 1`

(B) NY-R1(config)#`ip nat interface serial 0/0 overload`

(C) NY-R1(config)#`ip nat inside source list 1 interface serial 0/0 overload`

(D) NY-R1(config)#`ip nat inside source list 1`

259. What type of NAT maps a single public address to all internal addresses?

(A) Overloading

(B) Static

(C) Internal

(D) Public

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260. Your serial port is connected to the WAN environment, with your Fast Ethernet port connected to the LAN. When configuring NAT, how would you configure the Fast Ethernet port?

(A) NY-R1(config-if)#`ip nat outside`

(B) NY-R1(config-if)#`ip nat parallel`

(C) NY-R1(config-if)#`ip nat public`

(D) NY-R1(config-if)#`ip nat inside`

261. What command can you use to view the NAT translation table?

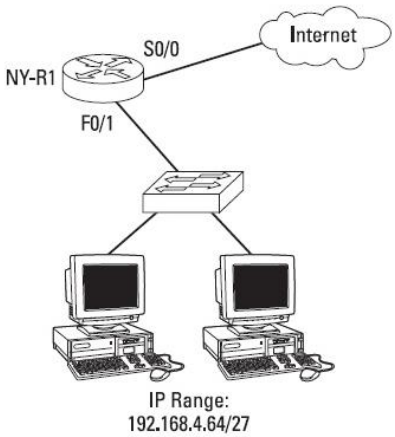
(A) `show ip nat`

(B) `show nat`

(C) `show nat translations`

(D) `show ip nat translations`

262. Looking at the figure below, which of the following would be used to configure NAT on the router? Assume the IP addresses are already assigned to the interfaces.



(A) Use the following commands:

NY-R1(config)#`Access-list 1 permit 192.168.4.64 255.255.255.224`

NY-R1(config)#`ip nat inside source list 1 interface serial 0/0 overload`

NY-R1(config)#`interface Serial0/0`

NY-R1(config-if)#`ip nat outside`

NY-R1(config-if)#`interface FastEthernet0/1`

NY-R1(config-if)#`ip nat inside`

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(B) Use the following commands:

```
NY-R1(config)#Access-list 1 permit 192.168.4.64 0.0.0.31
NY-R1(config)#ip nat inside source list 1 interface serial 0/0
overload
NY-R1(config)#interface Serial0/0
NY-R1(config-if)#ip nat inside
NY-R1(config-if)#interface FastEthernet0/1
NY-R1(config-if)#ip nat outside
```

(C) Use the following commands:

```
NY-R1(config)#Access-list 1 permit 192.168.4.64 0.0.0.31
NY-R1(config)#ip nat inside source list 1 interface Fast Ethernet
0/1 overload
NY-R1(config)#interface Serial0/0
NY-R1(config-if)#ip nat outside
NY-R1(config-if)#interface FastEthernet0/1
NY-R1(config-if)#ip nat inside
```

(D) Use the following commands:

```
NY-R1(config)#Access-list 1 permit 192.168.4.64 0.0.0.31
NY-R1(config)#ip nat inside source list 1 interface serial 0/0
overload
NY-R1(config)#interface Serial0/0
NY-R1(config-if)#ip nat outside
NY-R1(config-if)#interface FastEthernet0/1
NY-R1(config-if)#ip nat inside
```

Password Recovery

263–268 Choose the best answer(s).

263. What keystroke interrupts the boot sequence on a Cisco router in order to implement password-recovery procedures?

(A) Ctrl-Alt-Delete

(B) Ctrl-Break

(C) Fn-F4

(D) Fn-Break

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264. What is the default configuration register on most Cisco routers?

(A) 0x2142

(B) 0x2100

(C) 0x2102

(D) 0x2202

265. You wish to view the current configuration register value on a Cisco router, what command would you use?

(A) `show register`

(B) `show password`

(C) `show version`

(D) `show ip register`

266. After you have recovered the password on a Cisco device, what command would you use to set the configuration register back to the default?

(A) `config-register 0x2102`

(B) `config-register 0x2142`

(C) `config-register 0x2100`

(D) `config-register 0x1102`

267. When you need to recover a password, what bit number do you manipulate in the configuration register?

(A) 1

(B) 4

(C) 6

(D) 8

268. What is the new configuration register value after you configure it to bypass the loading of NVRAM?

(A) 0x2142

(B) 0x2100

(C) 0x2102

(D) 0x2202

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Access Control Lists (ACLs)

269–275 Choose the best answer(s).

269. Extended access lists use what range for access list numbers?

(A) 1-99

(B) 200-299

(C) 100-199

(D) 300-399

270. Your manager has asked you to block traffic from the system with the IP address of 192.168.5.100. You have configured an access list using the commands show in the figure below, but now no traffic from any system can pass through the interface. Why?

```
config term
access-list 1 deny host 192.168.5.100

interface s0/0
ip access-group 1 in
```

(A) Access lists have an implied deny all at the bottom.

(B) You need a subnet mask in the access list command.

(C) You should have used ip access-class instead of ip access-group.

(D) You should have used a permit instead of deny.

271. What type of access control list only allows you to filter traffic by the source IP address?

(A) Extended

(B) Standard

(C) Basic

(D) Advanced

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272. What type of access control list allows you to filter by source and destination IP address, source and destination port, and protocol?

(A) Advanced

(B) Basic

(C) Extended

(D) Standard

273. You would like to ensure that only systems on the 216.83.11.64/26 subnet can telnet into your router. What commands would you use?

(A) Use the following:

```
access-list 20 permit 216.83.11.64 0.0.0.63
line vty 0 4
access-class 20 in
```

(B) Use the following:

```
access-list 20 permit 216.83.11.64 255.255.255.192
line vty 0 4
access-class 20 in
```

(C) Use the following:

```
access-list 20 permit 216.83.11.64 0.0.0.63
line vty 0 4
access-group 20 in
```

(D) Use the following:

```
access-list 20 permit 216.83.11.64 255.255.255.192
line vty 0 4
access-group 20 in
```

274. You have created a standard access list # 20 that permits a group of IP addresses. You would like to use this access list to control who can telnet into the router. What command would you use on the telnet ports?

(A) `access-class 20 in`

(B) `access-group 20 in`

(C) `allow 20 in`

(D) `allow telnet 20 in`

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275. What is the result of the access list in the figure below?

```
access-list 100 deny ip host 192.168.10.50 192.168.2.0 0.0.0.255
access-list 100 deny tcp host 192.168.10.50 host 3.3.3.3 80
access-list 100 permit ip any any
```

(A) It denies the 192.168.10.50 system from accessing the 192.168.2.0 network, denies the system of 192.168.10.50 from accessing the website on 3.3.3.3, and permits all others.

(B) It denies the 192.168.2.0 network from accessing the 192.168.10.50 system, denies the system of 192.168.10.50 from accessing the website on 3.3.3.3, and permits all others.

(C) It denies the 192.168.10.50 system from accessing the 192.168.2.0 network, all systems from accessing the website on 3.3.3.3, and permits all others.

(D) It denies the 192.168.10.50 system from accessing the 192.168.2.0 network, all systems from accessing the website on 3.3.3.3, and denies all others.